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28 October 1976
SUPERSEDING
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21 March 1969

MILITARY SPECIFICATION

ALTIMETER, PRESSURE AAU-27/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of counter-drum-pointer, integrally lighted, pressure altimeter having a range of -1,000 to +50,000 feet, designated AAU-27/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-5087	Bonding, Electrical and Lightning Protection for Aerospace Systems
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

FSC 6610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-26500	Connectors, General Purpose, Electrical, Miniature, Circular, Environment Resisting, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MS27034	Connector, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS27573	Case, Instrument, 2-3/4 Dial, with Sump, Flange Mounted, Aircraft
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33649	Bosses, Fluid Connection, Internal Straight Thread

PUBLICATION

AFSC DESIGN HANDBOOK

DH 1-4	Electromagnetic Compatibility

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(Copies of specifications, standards, drawings and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity.

3.3.2 Protective treatment. When materials are used in the construction of the altimeter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.3 Toxic and corrosive fumes. The materials, as installed in the altimeter and under the conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.4 Design and construction. The altimeter shall be provided with a precision all-mechanical mechanism, a counter-drum-pointer altitude display, integral red and white lighting, and a barometric setting displayed by a 4-digit Veeder-type counter, or equal. The altimeter shall be so constructed that no parts will work loose in service. It shall be built to withstand normal shocks, vibrations, and other conditions incident to service, shipment, storage, and installation.

3.4.1 Maintainability. The maintenance concept for the altimeter shall be that it will be completely repairable at depot level only. The maintenance time shall not exceed 8 hours utilizing one depot technician and authorized depot AGE and technical information. Spare parts and facilities are assumed to be available. Maintenance time shall include all repair and test time.

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3.4.2 Reliability

3.4.2.1 Operational stability. The altimeter shall operate satisfactorily, continuously or intermittently, for a period equivalent to 1 year of flight and ground operation without any internal adjustment. The equivalent of 1 year is considered to be 1,000 hours of full-scale cycling with power applied to the vibrator and lighting circuits.

3.4.2.2 Mean-time-between-failures (MTBF). The altimeter shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level.

3.4.2.3 Longevity. The altimeter shall have a total operating life of 3 years (3,000 hours of full-scale cycling) with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.2.4 Shelf life. The altimeter shall remain operable over a shelf-life period of 2 years without internal adjustment. It shall exhibit stability and remain within calibration without zero setting adjustment over a period of 6 months.

3.4.3 Mechanism adjustment. The mechanism shall be provided with a means of adjusting or correcting the indications to conform to the performance requirements specified herein if adjustment is necessary during the life of the altimeter. The adjustment shall not affect the temperature performance of the instrument, shall be reasonably simple to manipulate, and shall be of such nature that adjustment can be accomplished by tools ordinarily used by an instrument repairman.

3.4.4 Barometric setting system. A setting system, controlled by a knob, shall be provided that will permit the altimeter display to be set to indicate zero altitude at any existing ground level pressure throughout the range of 28.10 to 31.00 (inclusive) inches Hg in 0.01-inch increments. The barometric scale shall be a 4-digit counter. When the barometric setting knob is rotated clockwise, it shall cause the scale to increase. The shaft of the setting knob shall withstand a load of 15 pounds applied perpendicular to the shaft at the extreme outward end of the knob without permanent deformation. A positive stop or stops shall be provided whereby the barometric zero setting mechanism cannot be moved beyond the range of 28.10 to 31.00 inches Hg. The stops shall withstand a torque of 6 pound-inches applied to the barometric setting knob without damage to the stops or instrument mechanism. The torque applied to the setting knob as normally required for setting barometric pressure shall be not more than 7 nor less than 2 ounce-inches throughout the setting range. The barometric

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counter unit displaying the setting shall be located as shown on figure 1. The mechanism shall withstand a pull of 15 pounds while the barometric scale is rotated throughout the range of settings from the minimum to the maximum stop without disengagement or damage of the mechanism.

3.4.4.1 Adjustment of barometric scale counter to pointer. The barometric setting system controlled by the setting knob in its extended position shall permit adjustment of the setting of the pointer to the barometric scale counter.

3.4.4.2 Setting knob position and lock. The barometric setting shall be provided with a lock which shall hold the setting knob in its normal position and allow adjustments as specified in 3.4.4. The means for unlocking the barometric setting system shall be accessible from the face of the altimeter when installed on the instrument panel. The unlocked barometric setting system shall allow adjustments as specified in 3.4.4.1.

3.4.5 Pointer, counter, and drum relationship. The pointer shall indicate readings in feet with reference to a single, circular, uniformly graduated scale making one revolution for each 1,000 feet of altitude. The pointer shall be positively geared to the 100-foot drum so that as the pointer moves cw (increasing altitude), the drum will rotate for increasing numbers synchronized with the pointer. The numerals on the 100-foot drum shall represent hundreds of feet. The 1,000-foot drum shall be driven by the 100-foot drum so that the 1,000-foot drum will index one digit as the 100-foot drum moves through the 9 to 0 position in either direction. During the long period between 0 and 9, the 1,000-foot drum shall be locked in position and in alignment with the 10,000-foot drum. Each digit on the 1,000-foot drum shall represent 1,000-foot intervals. The 10,000-foot drum shall be driven by the 1,000-foot drum so that the 10,000-foot drum will index one digit as the 1,000-foot drum moves through the short period between 9 and 0 in either direction. During the long period between 0 and 9 of the 1,000-foot drum, the 10,000-foot drum shall be locked in position. Each digit of the 10,000-foot drum shall represent 10,000-foot levels. In the space corresponding to zero, the 10,000-foot drum shall have striped lines, as shown on figure 2. The 1,000-foot drum and the 10,000-foot drum are collectively called the counter. The combined readings of the counter-drum shall indicate the pressure altitude in thousands and hundreds of feet. The pointer shall act as a vernier of the hundreds drum as well as being an indication of trend information. The counter-weight of the pointer shall not in any position obscure the indication of the drum. The display mechanism shall be such as to preclude high-frequency motions of the pointer or counters during normal operation.

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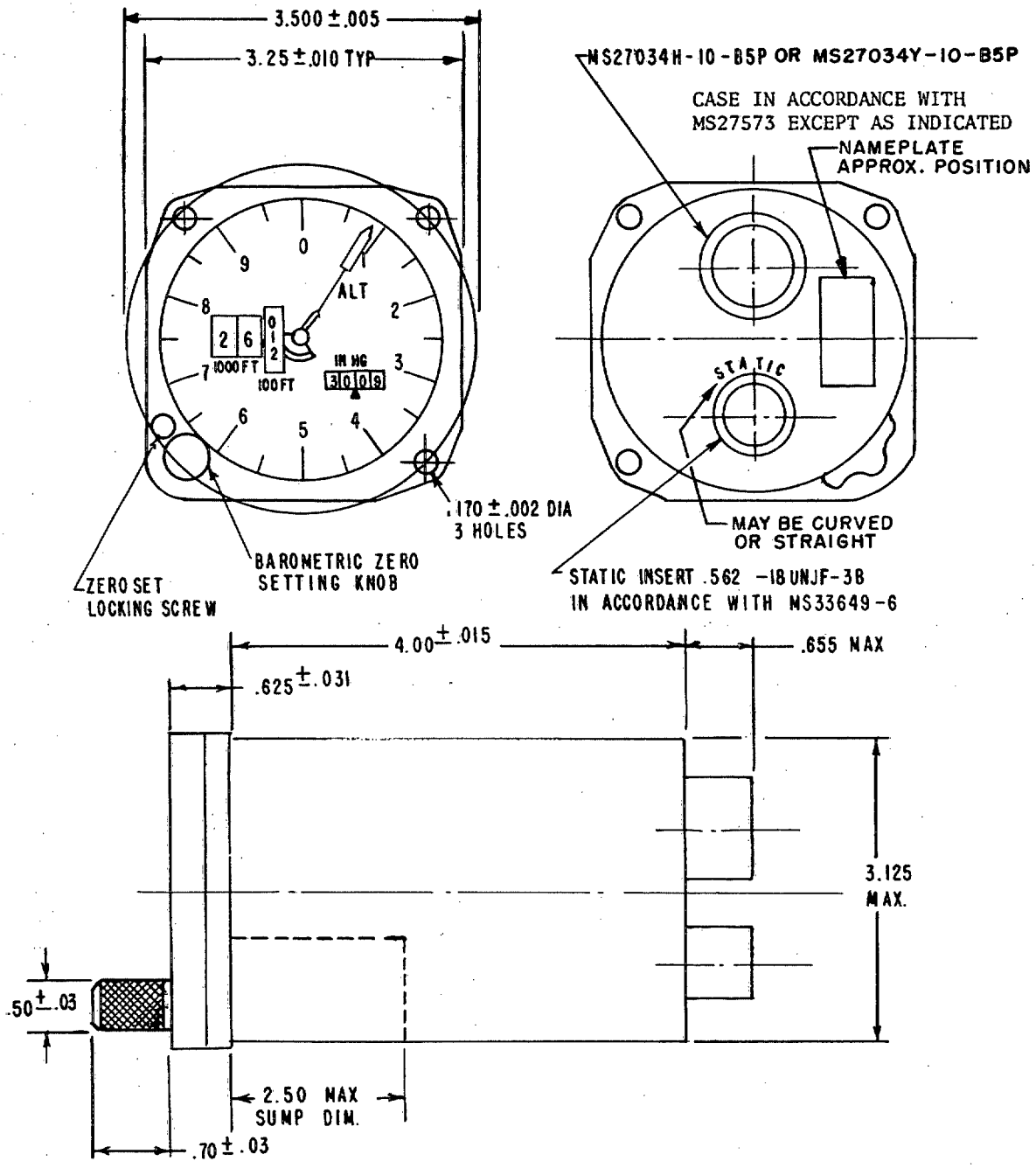


FIGURE 1. Case dimensions and dial.

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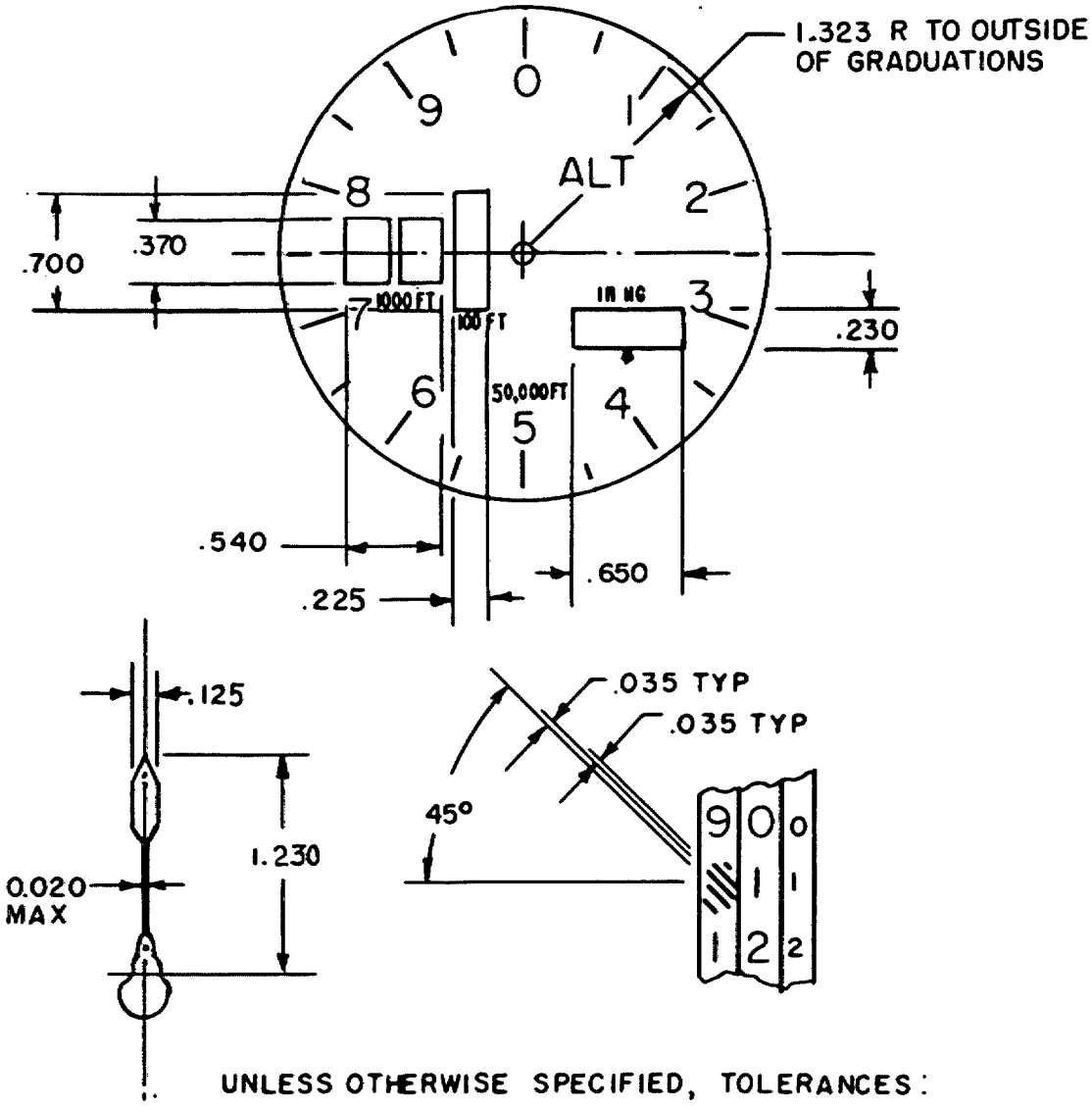


FIGURE 2. Dial.

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3.4.5.1 Pointer adjustment. The pointer and drum shall be free to swing to either side of their zero position without engaging a mechanical stop. The pointer shall be firmly attached to the pointer shaft to prevent slippage under normal service use and shall still be removable for depot service.

3.4.6 Case. The outline dimensions and the external connections of the altimeter case shall be as shown on figure 1. The body and mounting flange of the case shall be made of lightweight nonmagnetic metal uniform in texture. The body shall be finished in a durable black and the mounting flange shall be finished in durable lusterless black, color No. 37038 of FED-STD-595.

3.4.6.1 Static connection. The static connection shall be in accordance with figure 1 and shall be located as shown thereon. A 150-mesh wire filter screen (either stainless steel or monel) shall be firmly installed in the static connection to prevent dirt particles from entering the case. It shall be provided with a vented plug of acceptable material and design. Each plug shall incorporate a vent consisting of a hole .0625 ±.0156 inch in diameter.

3.4.7 Internal vibrator. An internal vibrator shall be incorporated into the altimeter which shall effectively decrease the friction in the mechanism. The vibrator shall be self-starting upon being energized, and shall start without being tapped or other external vibration assistance. The vibrator unit shall be capable of starting, running, and producing vibration sufficient to meet the friction and reliability requirements specified herein for a period of 1,000 hours without failure. The design shall be such that extended use of the vibrator will not result in deposits of foreign material nor produce deleterious fumes that may result in adverse effects on altimeter performance. The noise produced by the vibrator shall be held at a level below the ambient noise level of a cockpit during normal flight operations to prevent adverse effect on the pilot and crew operations.

3.4.7.1 Vibrator power circuit. The vibrator circuit shall be designed to operate on 28V dc power having characteristics in accordance with MIL-STD-704 for category B equipment. The input current shall not exceed 90 milliamperes.

3.4.8 Bonding. The altimeter shall be electrically bonded in accordance with MIL-B-5087.

3.5 Performance. The altimeter shall operate as specified herein after subjection to the following:

a. Temperatures: Continuous operation at temperatures ranging from -54° to +71°C and storage at temperatures ranging from -62° to +71°C

b. Humidity: Relative humidity up to 95 percent

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c. Altitude: Pressure altitudes ranging from -1,000 to +50,000 feet

d. Salt fog: Exposure to salt sea atmosphere

e. Vibration: Vibration as encountered in service use

f. Fungus: Fungus growth as encountered in tropical climates

g. Dust: Airborne sand and dust particles as encountered in desert areas

h. Shock: Shock of 15g for 11 ms (half sine shock pulse configuration)

i. Acceleration: Acceleration conditions as specified in 4.6.15.4

j. Overpressure: Pressures up to and including 35 inches Hg.

k. Underpressure: Pressures down to the equivalent of a 60,000-foot altitude.

3.5.1 Electromagnetic interference. Electromagnetic interference requirements shall be in accordance with MIL-STD-461 for class A1 equipment. DH 1-4 shall be used as a guide for designing the altimeter.

3.5.2 Magnetic effect. The altimeter, operating or nonoperating, shall not create a magnetic field with such intensity as to deflect a short bar magnetic compass more than 1°.

3.5.3 Dielectric strength. The altimeter shall withstand application of 500V 60 Hz for a period of 5 seconds.

3.5.4 Response. The altimeter shall track rates of change of altitude up to and including 50,000 fpm.

3.5.5 Accuracy. The performance of the altimeter shall be such that it will meet the tests and tolerances in Section 4 as well as any other parameter specified herein. The parameter tolerances are continuous and as such are linear between specified test points.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Electrical circuit

3.7.1 Electrical connector. The electrical connector shall be in accordance with MIL-C-26500, of the type shown on figure 1, and shall be connected as shown on figure 3. The connector shall be protected with a suitable cap.

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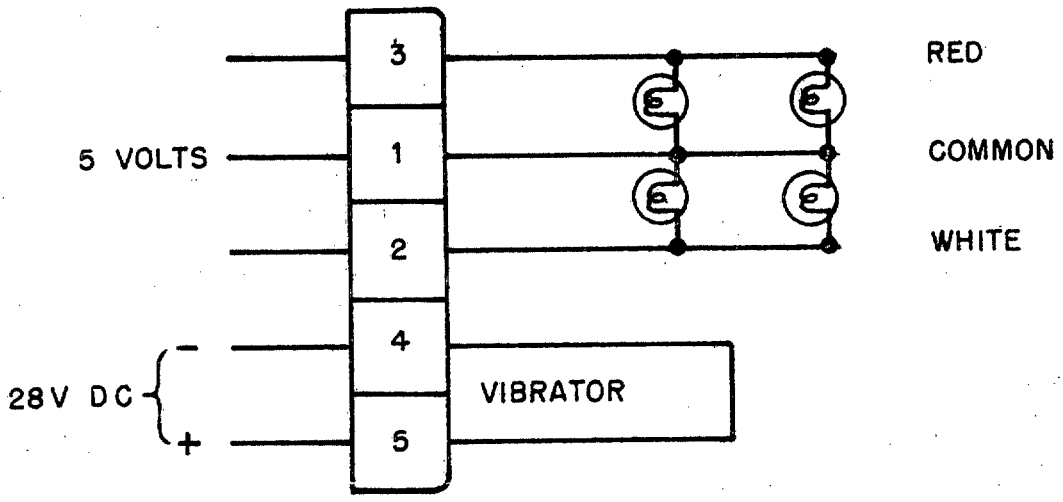


FIGURE 3. Electrical wiring.

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3.7.2 Overload protection. All parts and circuits of the altimeter which are likely to carry an overload due to any failures or improper adjustments shall be designed to withstand such overload without permanent damage. The use of fuses and other protective devices shall not be permitted.

3.7.3 Undervoltage protection. The altimeter shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage is within the specified limits.

3.7.4 Warmup time. The maximum warmup time for the equipment shall not exceed 1 minute under standard conditions nor 2 minutes at extreme service conditions.

3.7.5 Electrical ground. Electrical circuits shall not be grounded to the case.

3.8 Dial. The dial shall be marked as shown on figure 2. Unless otherwise specified, all visible portions of the dial shall be lusterless black, color No. 37038 of FED-STD-595, and all markings on the dial shall be lusterless white, color No. 37875 of FED-STD-595. The dimensions of the dial shall be as shown on figure 2. The dial shall be securely fastened to the mechanism frame.

3.8.1 Visibility of dial. The pointer, numerals, at least .0312 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 20° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.8.2 Markings. All markings shall be so durable as to withstand the usage encountered in service. The style and proportions of numeralsand letters placed on the dial shall conform to MS33558.

3.8.3 Numerals. Each numeral shall distinctly indicate the graduation to which it applies. If practicable, each numeral shall be so placed that the center mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. When several numerals are used in one group, the space between the numerals shall be approximately .0156 inch.

3.8.4 Finishes and dimensions. The following markings shall be finished in lusterless white, color No. 37875 of FED-STD-595. The dimensions shall be as specified.

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Markings	Height or Length Inch ±0.015	Width of Line or Graduation Inch ±0.005
Numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, on counters and 0, 1, 2, 3, 4, 5, 6, 9 on dial	0.281	0.035
Numerals 7, 8 on dial	0.187	0.023
100-foot graduation	0.202	0.030
50-foot graduation	0.109	0.015
Graduations for numerals 7 and 8	0.125	0.030
Baro counter numerals	0.156	0.025
ALT lettering	0.188	0.025
IN. Hg lettering	0.046	0.008
100 FEET lettering	0.046	0.008
1000 FEET lettering	0.046	0.008
Shaded portion of pointer	-----	-----
Border on altitude counter window	-----	0.035

3.8.4.1 The following markings shall be finished in lusterless black, color No. 37038 of FED-STD-595. The dimensions shall be as specified.

Markings	Inch ±0.015	Width of Line or Graduation Inch ±0.005
50,000 FEET lettering	0.062	0.011
Unshaded portion of pointer, dial, counters	-----	-----
Numerals 0 through 9 on 100 foot drum	0.156	0.025
AAU-27/A lettering	0.062	0.011

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3.8.4.2 The background of the 100-foot drum shall be finished in lusterless grey, color No. 36622 of FED-STD-595.

3.9 Integral lighting. The altimeter shall be provided with integral lighting in accordance with MIL-L-25467 and MIL-L-27160 to illuminate the display. The lighting circuit shall operate from 5V ac or dc and shall be connected as shown on figure 3. Power to the lighting circuit shall not exceed 4 watts.

3.9.1 Coverglass and lighting wedge. The coverglass and lighting wedge shall conform to MIL-G-81704, type I, and shall not interfere with the readability of the instrument. It shall be properly sealed and shall be replaceable by removing the bezel. All reflecting glass surfaces shall be provided with a reflection-reducing coating which shall conform to MIL-C-14806 and shall meet the environmental requirements specified herein.

3.9.2 Distance from dial to coverglass. The distance from the dial to the innermost glass surface shall not exceed 0.250 inch.

3.10 Components. The electric and mechanical components shall be in accordance with MIL-E-5400.

3.11 Jewel bearings. All sapphire and ruby jewel bearings used in the altimeter shall be in accordance with MIL-B-27497.

3.12 Threads. The threads of all machine screws shall conform to MIL-S-7742.

3.13 Finishes

3.14 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, and case inserts need not be anodized. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.15 Weight. The weight of the complete altimeter, excluding connecting fittings and mounting screws, shall not exceed 2.5 pounds.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The altimeter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

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3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the altimeter they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of altimeters shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard conditions. Unless otherwise specified, all tests shall be conducted under the following conditions:

Temperature:	Room ambient, 25° ±5°C
Pressure:	Normal atmospheric (approximately 29.92 inches Hg)
Humidity:	Room ambient, 40 to 90 percent relative humidity
Power:	5 ±0.1V ac or dc (lighting)
	28 ±1V dc (for integral vibration)

4.3.2 Standard pressures. Standard pressures used in calibrating and testing altimeters shall be in accordance with MIL-STD-859.

4.3.2.1 Reference standard barometer. The reference standard for atmospheric pressure shall be a mercury barometer which is maintained in accordance with the manufacturer's recommendations and is accurate (with corrections) within 0.005 inch traceable to the National Bureau of Standards. A reference barometer newly introduced into service shall be checked at intervals of approximately 6 months until the stability of its calibration has been established.

4.3.2.2 Barometer. During testing, a Type A-1 barometer with photoscanner or equivalent, properly calibrated for accuracy, shall be used to measure scale errors, hysteresis, and after effect. The barometer shall be properly maintained and checked in accordance with the handbook of instructions furnished by the barometer manufacturer. Altitude settings shall be in terms of the certified scale.

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4.3.3 Test readings. Unless otherwise specified, the integral vibrator shall be energized during all tests.

4.3.4 Attitude. Unless otherwise specified, the altimeter shall be tested in its normal operating position.

4.3.5 Integral lighting. The lighting system shall be energized and shall operate satisfactorily during all tests when an operational requirement for the altimeter is specified. The lighting shall be tested and shall operate satisfactorily after completion of each nonoperational test.

4.3.6 Rate of pressure change. Unless otherwise specified, the rate of change in pressure during all tests for both increasing and decreasing altitudes may be accomplished at any rate up to 6,000 feet per minute. To avoid passing the checkpoints, the rate shall be progressively reduced as the checkpoints are approached. The movement of the pointer during increasing and decreasing altitudes shall be smooth and free from irregular motion when the integral vibrator is operating.

4.3.7 Adjustments. Unless otherwise specified herein, no adjustments shall be made to the altimeter after initiation of testing. All sampling plan units and first articles shall contain a Government seal affixed prior to testing initiation. The seal to prevent adjustment shall remain intact unless otherwise authorized by Government personnel.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of 11 altimeters representative of the production equipment. The samples shall be identified with the manufacturer's own part number and such other information as required by the procuring activity. The tests shall be conducted on three groups of samples as follows:

a. Group 1 (three altimeters): All tests except reliability, longevity, and shelf life

b. Group 2 (six altimeters): Individual, Reliability, and Longevity

c. Group 3 (two altimeters): Individual and shelf life.

4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

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4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Barometric (setting scale, backlash, functional setting scale 1,500 feet, drum alignment after vibration)

c. Case leakage (room temperature, +45°C, -35°C)

d. Friction (stop and jump and static)

e. Position error

f. Scale error (room temperature, +45°C, -35°C)

g. Lighting (individual tests of MIL-L-25467 and MIL-L-27160).

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. Altimeters shall be selected at random in accordance with the following and subjected to the tests specified and as described under 4.6. Testing shall be completed prior to final shipment on the quantity specified and prior to shipment on any succeeding quantity.

Quantity Submitted for Acceptance	No. of Items to be Selected
First 15	1 (0 when sampling plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

a. Individual tests

b. Dielectric strength

c. Scale error at -54°C

d. Scale error at +71°C

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e. Vibration error

f. Lighting (sampling plan A tests of MIL-L-25467 and MIL-L-27160)

g. Response

h. Baro set strength test.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three altimeters shall be selected at random from the first 15 and three from each additional 500 produced on the contract or order and subjected to the following tests as described under 4.6. Testing shall be completed prior to final shipment on the quantity specified and prior to shipment on any succeeding quantity.

a. Sampling plan A tests

b. Low temperature exposure

c. High temperature exposure

d. Vibration failure

e. Acceleration

f. Shock

g. Humidity

h. Fungus

i. Salt fog

j. Dust

k. Overpressure

l. Underpressure

m. Magnetic property

n. Lighting (sampling plan B tests of MIL-L-25467 and MIL-L-27160)

o. Electromagnetic interference

p. Internal examination.

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4.5.2.3 Sampling plan C. Unless otherwise specified (see 6.2), five altimeters shall be selected at random from the first 15 of each quantity of 100 altimeters produced and subjected to the following tests as described under 4.6. The reliability testing for a quantity of 100 altimeters shall commence prior to Government acceptance of any altimeters of the succeeding quantity of 100 altimeters. No samples shall be selected from the last 2 months projected production quantities. Accept-reject criteria shall be as specified in 4.6.22.2.

a. Individual tests

b. Reliability tests.

4.5.2.4 Sampling plan D. One altimeter shall be selected at random from each monthly production quantity and subjected to the shelf life test specified in 4.6.22.4.

4.5.2.5 Sampling plan E. Unless otherwise specified (see 6.2), this plan is to be applied for Government acceptance of all submitted lots of altimeters on the contract. The sampling and inspection of specified lots or batches shall be in accordance with the latest issue of MIL-STD-105. General inspection Level II specified in MIL-STD-105, table I shall be used throughout the acceptance program. Samples drawn for inspection shall be randomly selected by the Government representative after the whole lot has been submitted for acceptance. Unless otherwise approved by the procuring activity, the lot size to be inspected shall be 30 altimeters except for the last lot of each month which may be between 26 and 50 altimeters inclusive. The acceptable quality level (AQL) to be maintained is 2.5. The sampling plan specified in MIL-STD-105, table IIA shall be used initially. The reduced sampling plan of table IIC may be used when the criteria for reduced sampling has been attained and with approval of the procuring activity. A tightened inspection shall not apply without approval of the procuring activity. Each sample selected shall be subjected to and shall meet all individual tests defined by 4.5.1 prior to acceptance of the lot by the Government.

4.5.2.5.1 Defective units. A defective unit is an altimeter which has not met all individual test requirements of this specification. The unit shall be considered defective as soon as it has failed an inspection criteria. Normal baroset adjustment (rezeroing) may be used to bring the unit within the requirements in which case it will not be considered defective.

4.5.2.5.2 Defective lots. Defective lots will be rejected and returned tc the contractor. The lot shall not be resubmitted for acceptance until, as a minimum, it has been completely reinspected by the contractor. Repairs shall be made as appropriate prior to complete retesting by the contractor.

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Samples inspected by the Government that are subject to and pass all inspection tests shall not be returned to the contractor with remainder of the defective lot but shall be accepted as a production item.

4.5.2.6 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.7 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in systems already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The altimeter shall be examined to determine compliance with the requirements specified herein with respect to workmanship, marking, general appearance, and the following:

a. Outline dimensions

b. Connectors

c. Alignment of counters in dial cutouts

d. Drum and pointer alignment shall be determined by causing the pointer to indicate 750 feet and visually checking drum numerals 7 and 8 to determine that they are equally spaced about the pointer shaft.

4.6.2 Barometric setting scale. The altimeter shall be subjected to atmospheric pressure or any controlled pressure. The pressure scale shall be set successively to the values specified in table I. After the series of readings has been recorded, the difference between the reading at the setting of 29.92 and each other setting shall be determined. These differences shall be recorded and compared with the values in the correct difference column of table I. The recorded values shall not differ from the correct differences by more than 25 feet at any setting. Atmospheric pressure shall be recorded before and after this test and proper allowance shall be made for any variation.

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TABLE I. Barometric setting scale.

Pressure Scale Setting	Correct Difference
28.10	-1,727
28.50	-1,340
29.00	-863
29.50	-392
29.92	0
30.50	531
30.90	893
31.00	983

4.6.2.1 Backlash. Immediately following the test specified in 4.6.2, the baro set scale shall be set successively to the values specified in table I starting at 31.0 and the pointer indication noted at each value. The difference between the indications noted and those specified in 4.6.2 shall not exceed 20 feet. Each indication shall be compared to the 29.92 indication specified in 4.6.2 and shall be within 25 feet of the correct difference of table I.

4.6.2.2 Baro set knot functional test. The baro setting system shall be operated from 28.10 to 31.00 and back to 28.10. The torque required for operation shall be measured and shall be within the values specified in 3.4.4.

4.6.2.3 Barometric setting scale 1,500 feet. Upon completion of tests required in 4.6.2, 4.6.2.1 and 4.6.2.2, the altimeter will be subjected to a decreasing pressure equivalent to 1,500 feet of altitude and held at that altitude. The pressure scale shall be set successively to the values specified in table I, and readings recorded. The difference between the reading at 29.92 inch Hg and all other readings shall be determined. These differences shall be recorded and compared with the values in the correct difference column of table I. The recorded values shall not differ from the values of table I by more than 25 feet at any setting. Allowance shall be made for any change in the set altitide that may occur during the performance of this test.

4.6.2.4 Barometric drum alignment after vibration. The barometric counter is aligned prior to vibration. After the internal vibrator has been operating for approximately 1 hour (±20 percent) all four original numerals of the barometric counter shall still be clearly visible in the dial cut-out. The setting of 29.92 inch Hg is to be used. This test may be performed as part of stop and jump friction testing with vibrator operating.

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4.6.3 Case leakage. This test shall be conducted at room temperature. A vacuum sufficient to produce a reading of approximately 30,500 feet shall be slowly applied to the static pressure connection of the altimeter at which point the connecting tubing shall be pinched off or otherwise completely sealed. The change in reading in 1 minute shall not exceed 100 feet. Altimeters shall be tested individually. This test may be combined with the tests in 4.6.6.

4.6.3.1 Case leakage at -35°C. The test specified in 4.6.3 shall be repeated except that 3 hours before and during this test the temperature shall be -35°C.

4.6.3.2 Case leakage at +45°C. The test specified in 4.6.3.1 shall be repeated except the temperature shall be +45°C.

4.6.4 Friction (stop and jump). The altimeter shall be tested for friction in both ascending and descending directions under the following conditions. With the internal vibrator operating, the altimeter shall be subjected to a constant rate decreasing and increasing pressure equivalent to 10,000 fpm. At 10,000 fpm and vibrator operative the change in reading of the pointer at the test points of table II, column A, due to its stopping and jumping as the counter is turning shall be recorded as friction and shall not exceed the tolerances specified in table II, column C. This test shall then be repeated at rates indicated in table II, column B, with vibrator inoperative. The friction shall be read and shall not exceed the tolerance specified in table II, column D.

TABLE II. Friction error.

Column A Test Point Altitude (Feet)	Column B Rate of Ascent or Descent (Feet/Min)	Column C Tolerance Vibrator Operative (Feet)	Column D Tolerance Vibrator Inoperative (Feet)
0	500	25	300
1,000	500	25	120
3,000	500	25	120
5,000	500	25	120
10,000	500	25	350
20,000	3,000	25	350
30,000	3,000	25	350
40,000	3,000	25	600

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4.6.4.1 Static friction. At the completion of the stop and jump friction test, the altimeter, with vibrator operating, shall be subjected to a constant rate of decreasing pressure not exceeding but approximately 3,000 fpm. This rate will continue until the controlled pressure approaches within the equivalent of 700 feet below the test point of table III being evaluated. At this time the rate will be decreased to 500 fpm ±50 fpm. The vibrator of the instrument shall be turned off as the instrument pointer passes through the 100-foot marking of the dial. The rate will again be reduced as the controlled pressure approaches within the equivalent of 200 feet below the test point of table III being evaluated. At this point the rate will be decreased to 300 fpm ±50 fpm. The pressure shall be stabilized at the desired value with no overshoot. This stabilized pressure shall agree with the table III value to within ±50 feet. The pressure shall be held constant at the stabilized value while two readings are taken. The first reading shall be taken with the vibrator inoperative and the second reading shall be made with the vibrator operating. The difference between the two readings shall be recorded as static friction and shall not exceed the tolerances listed in table III for each point.

TABLE III. Static friction error - vibrator inoperative.

Test Point Altitude (Feet)	Tolerance (Feet)
500	70
1,500	70
2,500	70
3,500	70
7,500	80
12,500	90
17,500	110
22,500	120
32,500	180
42,500	300

4.6.5 Position error. The altimeter position error readings shall be taken while the instrument is vented to atmosphere and in each of the following positions. The reading in any position shall not differ from the reading in the normal operating position by more than 20 feet.

a. Normal operating position

b. Rotated about the longitudinal axis 90°

c. Rotated about the longitudinal axis 180°

d. Rotated about the longitudinal axis 270°

e. Dial face up.

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4.6.6 Scale error

4.6.6.1 Scale error at room temperature. For a period of not less than 6 hours prior to this test, the altimeter shall not have been subjected to any test involving operation at other than atmospheric pressure. The altimeter shall be tested for scale errors as follows: The barometric setting shall be set at 29.92 inches Hg and shall remain at this setting during all scale error tests. Throughout the range, the altimeter shall be subjected to successively decreasing pressure and then increasing pressure as specified in table IV. The altimeter shall remain at the pressure corresponding to each test point for at least 1 minute but not more than 2 minutes before a test reading is taken. The scale error shall not exceed the tolerances specified in table IV. This test may be conducted with the instrument in an altitude chamber. When pressure is varied uniformly, the movement of the counter, drum, and pointer shall be free from backlash and any irregular motion.

4.6.6.1.1 Hysteresis and after effect. This test may be combined with the test specified in 4.6.6.1. Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to 50,000 feet in the test specified in 4.6.6.1, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding to the first test point in table V is reached. The altimeter shall remain at this pressure for at least 1 minute but not more than 15 minutes before the test reading is taken. The pressure shall again be increased at the same rate until the pressure corresponding to the second test point in table V is reached. The instrument shall remain at this pressure for at least 1 minute but not more than 10 minutes before the test reading is taken. The pressure shall again be increased at the same rate until 0 feet is reached. The reading of the instrument at the three test points shall not differ from the reading of the instrument for the same test points specified in 4.6.6.1 by more than the tolerance specified in table V. In addition, all test readings shall be within the tolerances of table IV.

4.6.6.2 Scale error at -35°C. The test specified in 4.6.6.1 shall be repeated except that for 3 hours before the test, with no power applied, and during the test the temperature shall be -35°C. The test shall be conducted within 2 minutes after power is applied. The test points and tolerances shall be in accordance with table IV.

4.6.6.3 Scale error at +45°C. The test specified in 4.6.6.2 shall be repeated except that the temperature shall be +45°C.

4.6.7 Lighting (individual tests of MIL-L-25467 and MIL-L-27160). The altimeter shall be subjected to the individual lighting tests of MIL-L-25467 and MIL-L-27160. These tests may be combined.

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TABLE IV. Scale errors.

Equivalent Pressure Units of Mercury at 0°C		Standard Altitude (Feet)	Scale Error Tolerance ±Feet		
MM 1/	Inches		-35°C	Room Temp and +45°C	-54°, +71°C
787.87	31.018	-1,000	45	30	
* 760.00	29.921	0	45	30	60
746.37	29.385	500	50	35	
732.93	28.856	1,000	50	35	
719.70	28.335	1,500	60	40	
* 706.65	27.821	2,000	60	40	80
693.80	27.315	2,500	70	45	
681.14	26.817	3,000	70	45	
656.38	25.842	4,000	75	50	
* 632.36	24.896	5,000	75	50	100
609.05	23.978	6,000	75	55	
564.51	22.224	8,000	80	60	
* 522.65	20.577	10,000	85	65	130
483.34	19.029	12,000	90	70	
446.46	17.577	14,000	95	75	
428.90	16.886	15,000	100	80	160
411.90	16.216	16,000	100	85	
379.53	14.942	18,000	105	90	
* 349.25	13.750	20,000	110	95	190
320.96	12.636	22,000	115	100	
282.03	11.103	25,000	120	105	210
225.69	8.885	30,000	135	115	230
* 178.83	7.041	35,000	150	125	
140.66	5.538	40,000	160	140	280
110.62	4.355	45,000	205	185	
* 86.99	3.425	50,000 2/	230	210	

1/ The asterisks in column 1 above denote test points after environmental tests (see 4.6.15).
2/ The 50,000-foot test point is for ascending altitude only.

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4.6.8 Dielectric strength. A voltage and frequency of 500V 60 Hz shall be applied between each terminal and the case for a period of 5 seconds. There shall be no insulation breakdown nor permanent damage to the altimeter as a result of this test. The voltage shall be raised from zero to maximum by rheostat control and not by switching.

4.6.9 Scale error at -54°C. The test specified under 4.6.6.1 shall be repeated except that for 3 hours before the test, with no power applied, and during the test the temperature shall be -54° ±2°C. The test shall be conducted within 2 minutes after power is applied. The test points and tolerances shall be in accordance with table IV.

4.6.10 Scale error at +71°C. The test specified in 4.6.9 shall be repeated except the temperature shall be +71° ±2°C.

TABLE V. Hysteresis.

Test Point	Standard Altitude (Ft)	Permissible Variation (Ft)
1st	25,000	60
2nd	20,000	60
3rd	0	30

4.6.11 Vibration error. The altimeter shall be mounted on the vibration stand in its normal operating position, secured by flange screws, and subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm so that any point on the case will describe a circular path inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the altimeter shall not occur in the specified frequency range. While the altimeter is being vibrated, the maximum amplitude of pointer oscillation shall not exceed 20 feet and pointer variation from the normal position shall not exceed 20 feet. The barometric setting shall not shift with respect to the setting prior to vibration. While the altimeter is being vibrated, the drum shall not oscillate more than 20 feet. Integral lighting shall be operating during this test and shall not flicker.

4.6.12 Lighting (sampling plan A of MIL-L-25467 and MIL-L-27160). The altimeter shall be subjected to the sampling plan A tests of MIL-L-25467 and MIL-L-27160. Those tests that are common may be combined.

4.6.13 Response. The altimeter shall be subjected to and shall track a pressure rate of change equivalent to 50,000 fpm. No damage to the internal mechanism nor zero or calibration shift shall result due to this test.

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4.6.14 Baro set strength test. The barometric setting knob shall be subjected to and shall meet the conditions specified in 3.4.4.

4.6.15 Environmental tests. Unless otherwise specified, the altimeter shall be subjected to the following tests conducted in accordance with the applicable procedures and methods of MIL-STD-810. Upon completion of each environmental test, the altimeter shall meet the tests specified in 4.6.3, 4.6.4, 4.6.6.1, and 4.6.7. When external connections are required to simulate installed conditions, they shall consist of AN and MS fittings and at least 10 feet of .25-inch copper tubing coiled and so arranged as to allow drainage of condensed vapors. They shall be attached to the static pressure port of the altimeter.

4.6.15.1 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with Method 502.1, procedure I. The storage temperature shall be -62°C.

4.6.15.2 High temperature exposure. The high temperature exposure test shall be conducted in accordance with Method 501.1, procedure I.

4.6.15.3 Vibration failure. The altimeter shall be subjected to a vibration failure test in accordance with Method 514.2, procedure I, equipment category (a), curve B.

4.6.15.4 Acceleration. The acceleration test shall be conducted in accordance with Method 513.2, procedures I and II, vehicle category airplanes. The forward acceleration factor (A) test levels shall be 2g. The test time duration in each direction shall be at least 1 minute following centrifuge stabilization.

4.6.15.5 Shock. The shock test shall be conducted in accordance with Method 516.2, procedure I, figure 516.2-2.

4.6.15.6 Humidity. The humidity test shall be conducted in accordance with Method 507.1, procedure I. The altimeter shall be tested within 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion and deterioration that would affect subsequent operation of the instrument.

4.6.15.7 Fungus. The fungus test shall be conducted in accordance with Method 508.1, procedure I. At the end of the test period, the altimeter shall be thoroughly examined to ascertain that no fungus growth has occurred.

4.6.15.8 Salt fog. The salt fog test shall be conducted in accordance with Method 509.1, procedure I, for a period of 50 hours. There shall be no corrosion or other damage as a result of this test that would affect subsequent operation of the altimeter.

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4.6.15.9 Dust (fine sand). The dust test shall be conducted in accordance with Method 510.1, procedure I.

4.6.16 Overpressure. With the barometric setting set at 29.92 inch Hg, a pressure of 35 inches Hg absolute shall be applied to the static fitting of the altimeter. There shall be no damage to the internal mechanism due to pressure and no zero or calibration shift.

4.6.17 Underpressure. The altimeter shall be subjected to a pressure equivalent to an altitude of 60,000 feet. There shall be no damage to the internal mechanism of the altimeter and no zero or calibration shift.

4.6.18 Magnetic property. The altimeter, not energized, shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the altimeter 5.5 inches from and magnetically East or West of the center of the compass. Starting directly under the compass, the altimeter shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the altimeter shall be rotated about on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. With the altimeter at any specified position, the compass deflection shall not exceed 1°. This test shall be repeated with the altimeter energized with rated voltage.

4.6.19 Lighting (sampling plan B tests of MIL-L-25467 and MIL-L-27160). The altimeter shall be subjected to the sampling plan B tests of MIL-L-25467 and MIL-L-27160. Those tests which are common may be combined.

4.6.20 Electromagnetic interference. The altimeter shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, methods CE03, CS01, CS02, CS06, RE02, and RS01.

4.6.21 Internal examination. The case of the altimeter shall be opened and the internal mechanism examined to determine compliance with the applicable requirements specified herein. The lamp circuits shall also be examined to determine compliance with MIL-L-25467 and MIL-L-27160. Any damage or deterioration to the mechanism as a result of preceding tests or any deficiency that would in any way prevent the altimeter from meeting the requirements of this specification shall be cause for rejection.

4.6.22 Reliability

4.6.22.1 Preproduction. The preproduction reliability test shall be conducted in accordance with MIL-STD-781, test level E, which shall demonstrate an altimeter minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level. The altimeter shall be pneumatically cycled from minimum to maximum altitude

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and return to minimum (one cycle) within approximately 15 minutes for a total of at least four cycles per hour. During the external vibration period (10 minutes of each hour), the cycling rate shall be increased to 20,000 fpm. Period B shall be approximately 4 hours in duration. During the cooling period, the power shall be turned off and the altitude input returned to sea level. The baro set knob shall be rotated once throughout the complete range each 24-hour period. In case of malfunction, the malfunction shall be recorded, the test interrupted while repairs are made, and the test then continued from the point of interruption. At approximately 200-hour intervals, the altimeter shall be subjected to the tests specified in 4.6.2, 4.6.2.1, 4.6.2.2, 4.6.3, 4.6.4, 4.6.4.1, 4.6.6.1, and 4.6.6.1.1. At the conclution of this test, the altimeter shall meet all individual tests. If recalibration, adjustment, or repair is required to pass any tests, a failure shall be charged against this test. A 25-foot out-of-tolerance condition in scale error will be permitted provided rezeroing by use of the baro set system will correct the condition. This rezeroing shall not result in any out-of-tolerance condition. Scale error tests at all test temperatures shall be performed after rezeroing. Rezeroing will be allowed only once per instrument per test. During all cycling, the vibrator shall be operating and any failure shall be charged as a malfunction against this test. The lighting system shall also be operating during cycling and upon completion of the final cycle, the altimeter shall meet the lighting tests specified in 4.6.7, 4.6.12, and 4.6.19. Failure to meet these tests shall be considered a malfunction. Time accrued during operation of the altimeter in previous tests may be counted as part of the time required by this test. This test may be interrupted for normal weekend shutdown. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan III. If five or more failures occur, the altimeter will have failed the test.

4.6.22.2 Production. The production reliability test shall be conducted in accordance with 4.6.22.1 except accept-reject criteria shall be in accordance with MIL-STD-781, test plan VI.

4.6.22.3 Longevity. Two of the six preproduction reliability test samples shall be subjected to a longevity test in accordance with MIL-STD-781, test plan XXVIII.

4.6.22.4 Shelf life. The altimeter shall be placed in storage for a minimum period of 30 days. During the 30-day period the altimeters shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by the normal atmospheric pressure variations. The temperature shall remain constant; however, normal room temperature variations will be permitted. At the end of the 30-day-period, the altimeters shall meet the scale error at room temperature, friction (stop and jump), and static friction tests.

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4.6.23 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471. The tasks shall be selected in advance of timing such tasks and shall be approved by the procuring activity.

4.7 Inspection of preparation for delivery. This inspection shall be conducted in accordance with the quality assurance provisions of MIL-STD-794 to insure compliance with Section 5.

5. PACKAGING

5.1 Preservation-packaging. Unless otherwise specified in the individual contract or order, the following shall apply.

5.1.1 Level A (reference definition in MIL-STD-794). Water-vaporproof caps or plugs shall be applied to electrical connectors and other openings of the altimeters. Each unit shall be packaged in accordance with MIL-P-116, method IC-1, one each in a unit container in accordance with PPP-B-636, domestic class. Cushioning, thickness, density, and configuration shall insure shock transmission does not exceed peak values in g's established for the altimeter when complete packs are subjected to the rough handling drop tests stipulated by MIL-STD-794.

5.1.2 Level C (reference definition in MIL-STD-794). Altimeters shall be cushioned identical to level A requirements specified in 5.1.1. Water-vaporproof caps or plugs, waterproof heat-sealed barriers, and desiccating agents shall be omitted. Boxes in accordance with PPP-B-636, domestic class, shall be used as the unit container.

5.2 Packing. Unless otherwise specified in the individual contract or order, the following shall apply.

5.2.1 Level A. Altimeters packaged as specified in 5.1.1 shall be packed in shipping containers in accordance with PPP-B-636, weather-resistant class. Shipping containers shall be of uniform shape and size and of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Altimeters packaged as specified in 5.1.1 shall be packed in boxes in accordance with PPP-B-636, domestic class. Other requirements specified in 5.2.1 shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirement of this level.

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5.3 Marking. Unless otherwise specified in the individual contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The AAU-27/A altimeter covered by this specification is intended for use in aircraft to indicate the pressure altitude of the aircraft above smoe reference point assuming standard conditions of temperature and pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted (see 4.5.2.2)

c. When sampling plan C tests will not be conducted (see 4.5.2.3)

d. Levels of packaging and packing required (see section 5)

e. Which sampling plan tests (ie, Sampling plan A or Sampling plan E) will be required.

6.3 International standardization agreement. Certain provisions (see figure 1) of this specification are the subject of international standardization agreements STANAG 3319 and ASCC Air Standard 10/7. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreements concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Air Force - 11

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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Check the source to verify that this is the current version before use.